76 Myelodysplastic/ Myeloproliferative Neoplasms

Francesco Onida and Yves Chalandon

76.1 Definition, Epidemiology, Diagnosis, and Classification

The myelodysplastic-myeloproliferative neoplasms (MDS/MPNs) are a heterogeneous group of hematologic malignancies characterized by dysplastic and myeloproliferative clinical, laboratory, and morphological overlapping features, both in marrow and in blood. The MDS/MPN category, recently updated by the last revision to the WHO classification of myeloid and histiocytic/dendritic neoplasms (Khoury et al. 2022), includes chronic myelomonocytic leukemia (CMML), MDS/MPN with neutrophilia (previously called atypical CML), MDS/MPN with SF3B1 mutation (in its absence with ringed sideroblasts) and thrombocytosis (MDS/MPN-SF3B1-T), as well as MDS/MPN not otherwise specified (MDS/MPN-NOS) (Table 76.1).

MDS/MPN are typically diagnosed in elderly age with CMML being definitely the most frequent subtype (incidence of around 0.3–0.4 case/100,000 inhabitants per year, median age 74–79 years) (Benzarti et al. 2019). Being very uncommon, data concerning the incidence of MDS/MPN with neutrophilia, MDS/MPN-SF3B1-T, and MDS/MPN-NOS are currently unknown.

F. Onida (✉)
Hematology, ASST-Fatebenefratelli-Sacco, University of Milan, Milan, Italy
e-mail: francesco.onida@unimi.it

Y. Chalandon
Hematology Division, Department of Oncology, Hôpitaux Universitaires de Genève, University of Geneva, Geneva, Switzerland
e-mail: yves.chalandon@hcuge.ch

© The Author(s) 2024
A. Sureda et al. (eds.), *The EBMT Handbook*, https://doi.org/10.1007/978-3-031-44080-9_76

Table 76.1 Classification and diagnostic criteria of MDS/MPNs

Disease	Blood findings	Bone marrow findings
Chronic myelomonocytic leukemia (CMML): Myelodysplastic CMML (MD-CMML) Myeloproliferative CMML (MP-CMML)	Monocytes ≥0.5 × 10^9/L[a], and relative (≥10%) <20% blasts MD-CMML: <13 × 10^9/L WBC MP-CMML: ≥13 × 10^9/L WBC	<20% blasts[b] Not meeting the criteria of chronic myeloid leukemia or other myeloid neoplasms. Not meeting diagnostic criteria of myeloid/lymphoid neoplasms with tyrosine kinase fusions. Dysplasia in one or more myeloid lineages[c]
Myelodysplastic/ myeloproliferative Neoplasms with neutron-philia (previously called atypical CML)	Leukocytosis due to increased numbers of neutrophils with IMC, ≥10% of WBC Basophils <2%, Monocytes <10% <20% blasts Dysgranulopoiesis	<20% blasts Dysgranulopoiesis Hypercellularity, with granulocytic proliferation and granulocytic dysplasia, ±dysplasia in the erythroid and megakaryocytic lineages No evidence of BCR/ABL1, PDGFRA, PDGFRB, or FGFR1 rearrangement or PCM1-JAK2
MDS/MPN with SF3B1 mutation and thrombocytosis	Anemia, ≥15% ring sideroblasts, platelet count ≥450 × 10^9/L <1% blasts	Erythroid lineage dysplasia with or without multilineage dysplasia, <5% blasts The presence of a *SF3B1* mutation or, in its absence, no history of recent cytotoxic or growth factor therapy that could explain the MD/MP features and presence of ≥15% ring sideroblast Then called MDD/MPN with ring sideroblasts and Thrombocytosis
MDS/MPN not otherwise specified (MDS/ MPN-NOS)[d]	<20% blasts	<20% blasts

IMC immature myeloid cells (promyelocytes, myelocytes, and metamyelocytes)
[a]If blood monocytes ≥0.5 but <1.0 × 10^9/L, acquired clonal cytogenetic or molecular abnormality is an additional required criteria for CMML diagnosis. Abnormal partitioning of peripheral blood monocyte subsets is introduced as a new supporting criterion
[b]Including blasts equivalent: myeloblasts, monoblasts, and promonocytes
[c]Morphologic dysplasia should be present in ≥10% of cells of a haematopoietic lineage in the bone marrow
[d]Myeloid neoplasms with mixed proliferative and dysplastic features that do not meet the criteria for CMML, MDS/MPN with neutrophilia, or MDS/MPN-with SF3B1 mutation and thrombocytosis are classified as MDS/MPN-NOS

76.2 Risk Factors and Prognostic Index

The clinical course of MDS/MPN varies from an indolent course over several years for a minor fraction of patients with CMML and MDS/MPN-SF3B1-T to a more rapid progression with dismal prognosis and frequent transformation into secondary acute myeloid leukemia in the preponderance of patients with CMML and in the vast majority of patients with MDS/MPN with neutrophilia and MDS/MPN-NOS, for whom allo-HCT still represents the only curative option (Onida and Beran 2008; Onida 2017; Patnaik and Tefferi 2022, 2023).

CMML is highly heterogeneous, with clinical and hematological characteristics varying from mainly myelodysplastic to predominantly myeloproliferative which has led both the last version of the WHO classification (Khoury et al. 2022) and the International Consensus Classification (Arber et al. 2022) to retain the distinction of MD-CMML and MP-CMML variants, on the base of total WBC count < or ≥13 × 10^9/L, respectively (Bennett et al. 1994). Based on marrow and peripheral blood blast percentage, there are now only two disease subtypes (CMML-1, and CMML-2), associated to a corresponding decreasing life expectations (Khoury et al. 2022; Arber et al. 2022). Over the latest

Table 76.2 Prognostic systems in MDS/MPN

MDS/MPN	Prognostic model	Variables included [score]	Risk groups	Median OS (months)
CMML	GFM (Itzykson et al. 2013)	Age > 65 years [2] WBC >15 × 10^9/L [3] PLT <100 × 10^9/L [2] Anemia (F < 10 g/dL, M < 11 g/dL) [2] *ASXL1* mutation [2]	Low (score ≤ 4) Intermediate (score = 5–7) High (score ≥ 8)	Not reached 38.5 14.4
	CPSS (Such et al. 2013)	WBC ≥13 × 10^9/L [1] WHO CMML-2 subtype [1] CMML-specific CG[a] inter/high [1/2] Transfusional dependency [1]	Low (score = 0) Intermediate-1 (score = 1) Intermediate-2 (score = 2–3) High (score = 4–5)	72 31 13 5
	MMM (Patnaik et al. 2014)	Hb <10 g/dL [2] AMC >10 × 10^9/L [2] Circulating IMC >0% [2] PLT <100 × 10^9/L [1.5] *ASXL-1* Mut (frameshift/nonsense) [1.5]	Low (score = 0) Intermediate-1 (score = 1.5–2) Intermediate-2 (score = 2.5–4.5) High (score ≥ 5)	97 59 31 16
	CPSS-Mol (Elena et al. 2016)	WBC ≥13 × 10^9/L [1] BM blasts ≥5% [1] Genetic risk group[b] [score 0–3] Transfusional dependency [1]	Low (score = 0) Intermediate-1 (score = 1) Intermediate-2 (score = 2–3) High (score ≥ 4)	Not reached 64 37 18
MDS/MPN with neutrophilia	MDACC (Onida et al. 2002)	Age > 65 years [1] Hb ≤10 g/dL [1] WBC >50 × 10^9/L [1]	Low (score = 0–1) High (score = 2–3)	38 9
MDS/MPN-SF3B1-T	Mayo (Patnaik et al. 2016)	Hb <10 g/dL [1] CG abnormalities [2] ASXL-1 mutation [1] SETBP1 mutation [1]	Low (score = 0) Intermediate (score = 1) High (score ≥ 2)	80 42 11

[a]CMML-specific cytogenetic risk classification, low, normal, and isolated –Y; intermediate, other abnormalities; and high, trisomy 8, complex karyotype (≥3 abnormalities), and abnormalities of chromosome 7

[b]Genetic risk group, CMML-specific cytogenetic risk classification + *ASXL1/NRAS/SETBP1* mutation (score = 1)/*RUNX1* mutation (score = 2)

years, a number of disease-specific prognostic systems have been developed in CMML in order to allow the best treatment strategy allocation (Onida 2017). The most recent ones are listed in Table 76.2.

MDS/MPN with neutrophilia, previously called atypical CML, is a rare hematologic malignancy with an overall dismal prognosis (median 24 months). Age, hemoglobin level, and leukocyte count have been identified as variables with independent prognostic significance, allowing the stratification of two groups with significantly different life expectations. Likewise, for MDS/MPN-SF3B1-T, three risk categories of patients were recently differentiated by a Mayo Clinic prognostic model including molecular investigations (Table 76.2).

MDS/MPN-NOS is the most heterogeneous and the least well-characterized entity, with no currently recognized specific molecular findings. Some description of the biological and clinical characteristics have been reported in two series (DiNardo et al. 2014; Wang et al. 2014), with median survival of 12.4 and 21.8 months, respectively, and possible association of thrombocytosis with a more favorable outcome. More recently, a

retrospective analysis including 88 patients from the Mayo Clinic and 47 from the Moffitt Cancer Center reported median OS of 26 and 33 months, respectively (Mangaonkar et al. 2020).

76.3 Pretransplantation Treatment

For this rare group of diseases, there are few prospective studies on therapy, most being either retrospective analyses or case reports, making it difficult to give recommendations. Similarly to MDS, in high-risk patients with MDS/MPN and low blast count (<10%), upfront transplantation is the most frequently preferred strategy. In general, because apart from allo-HCT, no therapy has been shown to modify the disease course, pretransplantation treatments point toward symptom control rather than the achievement of disease remission (Odenike et al. 2015).

76.3.1 CMML

In general, treatment strategies in patients with CMML with symptomatic or progressive disease are based on the dysplastic versus proliferative features and the percentage of marrow blasts (Onida et al. 2013; Itzykson et al. 2018). In the presence of rising leukocytosis and/or organ infiltration (mostly splenomegaly) with low marrow blast percentage, hydroxyurea (HU) remains the drug of choice. Patients showing high blast percentages may be bridged to transplant through AML-like induction chemotherapy or by means of hypomethylating agents (HMAs), with a reported 40–50% overall response rate (Patnaik and How 2022). New treatment strategies based on the combination of HMAs with other agents (e.g., lenalidomide, venetoclax) or innovative modalities including targeted therapies (e.g., JAK2 inhibitors, MEK1/2 inhibitors, IDH1/2 inhibitors) or receptor agonists/antagonists immunotherapy approaches currently under experimental phase (Renneville et al. 2021) may further increase the response rate leading to an overall improvement of post-transplantation outcomes.

76.3.2 MDS/MPN with Neutrophilia (Previously known as Atypical CML)

Due to its absolute rarity in patients having no age or comorbidity barrier to allo-HCT, no consensus subsists on to whether any pretransplant treatment may have an impact on post-transplantation outcome and what kind of therapy should be best used. Control of leukocytosis is generally achieved with cytoreductive agents such as HU or IFN-α immunomodulation. Chemotherapy induction treatment is preferred when facing high blast count in advanced disease phases or in patients showing AML transformation.

Some efficacy of decitabine (Tong et al. 2015) and of ruxolitinib single agent (Dao et al. 2020) has also been reported, whereas a phase II trial of AZA and ruxolitinib in combination in a series of 35 MDS/MPN patients showed promising activity, with an overall response rate of 57% according to the 2015 international consortium response criteria for MDS/MPN (Savona et al. 2015), even though median survival of the few aCML included patients (n = 4) was only 8 months (Assi et al. 2018). According to recently reported data, ASXL1, *SETBP1,* and *ETNK1* are the most frequently mutated genes, with a total of 43.2%, 29.7, and 16.2%, respectively (Fontana et al. Hemasphere 2020), whereas *JAK2* mutation is rare (0–7%), and *CSF3R* mutations are only occasionally observed (Sun et al. 2023). Even though in future these findings may influence therapeutic approaches by means of evolving targeted therapies, currently allo-HCT remains the only treatment strategy with established curative potential in eligible patients (Dao et al. 2017; Patnaik and Tefferi 2023).

76.3.3 MDS/MPN-SF3B1-T

MDS/MPN-SF3B1-T generally represents the disease entity associated with the best prognosis among overlap syndromes, with a median survival of about 6 years (Kuendgen et al. 2021). Guidelines for disease management are not formally recognized, and treatment strategies are generally extrapolated from low-risk MDS and MPN, with adjusted individual management

depending on presenting problems (Patnaik and Tefferi 2017). While lenalidomide has been occasionally reported to reduce transfusion need, antiplatelet, and cytoreductive treatments are often required due to the high risk of thrombosis (Kuendgen et al. 2021). Similar to lower risk MDS and MPN patients, allo-HCT is reserved for patients with refractory cytopenias, high-risk cytogenetics and/or molecular genetics, or progressive disease (Patnaik et al. 2016). Based on the different gene mutations possibly involved (*SF3B1 in ~85%, JAK2V617F in ~50%, TET2 in ~25%, ASXL1 in ~20%, DNMT3A in ~15%, and SETBP1 in ~10%.*), attentiveness in targeted therapies is developing.

76.3.4 MDS/MPN-NOS

MDS/MPN-NOS is a very rare and heterogeneous disease entity, with no consensus on which therapy (if any) should be given for patients candidate to allo-HCT. Allogeneic HCT should be encouraged in eligible patients with TP53 mutations (Patnaik and Tefferi 2023). Augmented leukocyte proliferation is generally managed by means of cytoreductive agents such as HU or through immunomodulation with IFNα, while HMAs as well as lenalidomide may represent an option in case of prevailing cytopenias. JAK inhibitors are also potential therapeutic options, either alone or in combination with HMAs (Assi et al. 2018). When patients are progressing to AML transformation, induction chemotherapy should be used as a bridge to allo-HCT.

76.4 Autologous HCT

Because the harvesting of polyclonal hematopoietic progenitor cells is not feasible through the currently available treatment options, autologous HCT is currently not a recommended strategy in MDS/MPN.

76.5 Allogeneic HCT

Currently still representing the only curative strategy, the role of allo-HCT in adult MDS/MPN patients remains controversial mainly due to the lack of prospective studies, being therefore generally considered a possible treatment option for eligible patients with high-risk diseases.

In *CMML* benefits and risks of allo-HCT have been analyzed retrospectively in various series, with different characteristics at transplant and much variable outcomes described (Table 76.3). Recent recommendations from an international expert panel agreed to limit indication for allo-HCT in CMML patients classified in the intermediate-2 and high-risk CPSS categories (de Witte et al. 2017), representing the preferred treatment modality for younger patients with acceptable comorbidity index (Patnaik et al. 2015; Patnaik and Tefferi 2022). This indication was recently reinforced by the results of a study comparing transplant vs no transplant strategy in a large series of CMML patients from an international collaborative analysis (Robin et al. 2022).

As *MDS/MPN with neutrophilia* is extremely rare in people younger than 65 years, outcome after allo-HCT has been described only in small single-institution series. A 5-years OS and RFS of 51% and 36%, respectively, were recently reported by the EBMT-CMWP in a retrospective analysis of 42 patients transplanted between 1997 and 2006. With an RR of 40%, a better OS was recognized in young patients with low EBMT risk score (Onida et al. 2017).

In *MDS/MPN-SF3B1-T* allo-HCT is generally not indicated, being reserved for patients developing refractory cytopenias or accelerated/blastic transformation (Sharma et al. 2017), whereas eligible patients with MDS/MPN-NOS should always be considerate as potential candidate for allo-HCT due to the general dismal prognosis (Mangaonkar et al. 2020).

Table 76.3 Summary of selected studies on allo-HCT in CMML

Author (year)	Pt N.	Median age (range)	Disease type/stage	Donor type	Conditioning (MAC vs RIC)	TRM/relapse rate	Survival outcome
Kröger et al. (2002)	50	44 (19–61)	CMML-1 = 28 CMML-2 = 17 Missing = 5	MRD = 43 MUD = 7	MAC = 50 RIC = 0	TRM = 52% RR = 28%	OS (5y) = 21% DFS (5y) = 18%
Eissa et al. (2011)	85	51 (1–69)	CMML-1 = 57 CMML-2 = 26	MRD = 38 MUD = 47	MAC = 58 RIC = 27	TRM (10y) = 35% RR (10y) = 27%	OS (10y) = 40% DFS (10y) = 40%
Park et al. (2013)	73	53 (27–66)	CMML-1 = 24 CMML-2 = 26 Missing = 23	MRD = 41 MUD = 32	MAC = 30 RIC = 43	TRM = 35% RR = 35%	OS (3y) = 32% DFS (3y) = 29%
Symeonidis et al. (2015)	513	53 (18–75)	CMML-1 = 87 CMML-2 = 32 s-AML = 95 Missing = 299	MRD = 285 MUD = 228	MAC = 249 RIC = 226	TRM (4y) = 41% RR (4y) = 32%	OS (4y) = 33% DFS (4y) = 27%
Kongtim et al. (2016)	83	57 (18–78)	CMML-1/2 = 47 sAML = 36	MRD = 30 MUD = 47 MMR = 6	MAC = 64 RIC = 19	TRM (3y) = 31% RR (3y) 33%	OS (3y) = 35% DFS (3y) = 34%
Liu et al. (2017)	209	57 (23–74)	CMML-1 = 140 CMML-2 = 52 Missing = 17	MRD = 73 MUD = 127 MMUD = 9	MAC = 105 RIC = 99 Missing = 5	TRM (5y) = 28% RR (5y) = 52%	OS (5y) = 30% DFS (5y) = 20%
Itonaga et al. (2018)	159	54 (16–75)	Not reported	MRD = 51 MUD (BM) = 66 Cord = 30 MMR = 12 MDR = 29	MAC = 92 RIC = 67	TRM = 28% RR = 39%	OS (3y) = 33%
Pophali et al. (2020)	70	58 (18–73)	CP 66% BT 34%	MUD = 32 mMUD/haplo = 9	MAC = 32 RIC = 38	TRM = 28% RR = 27%	OS (5y) = 51% (CP) OS (5y) = 19% (BT)
Rovó et al. (2023)	1466	60 (IQR 54–65)	CMML-1 = 52% CMML-2 = 48% Missing = 68%	MRD = 28% MUD/mMUD = 64% Haplo = 8%	MAC = 36% RIC = 64%	TRM = 32% RR = 37%	OS (5y) = 37% PFS (5y) = 31%

76.6 Source of HSC

No impact of HSC source on the transplant outcome has been observed in the largest CMML series reported by the EBMT-CMWP (Symeonidis et al. 2015). This was in contrast to the CIBTMR study, in which the survival was statistically better with PBMC than with BM, with no clear explanation outside the small proportion of BM transplants (16%) (Liu et al. 2017). The source of stem cell is therefore left open, but PBSC may potentially be preferred to decrease the risk of graft failure and the relapse risk, particularly with the use of RIC.

For MDS/MPN with neutrophilia, MDS/MPN-SF3B1-T, and MDS/MN-NOS, data are too scarce to make clear recommendations.

76.7 Conditioning and GvHD Prophylaxis

In MDS/MPN patients, the choice of conditioning regimen depends on many different conditions, the major ones being comorbidities, patient age, disease phase at transplant, type of donor, and HSC source. In the two largest retrospective series of CMML patients (Symeonidis et al. 2015; Liu et al. 2017), MAC and RIC were almost equal in proportion, with no outcome difference. Likewise, in the largest reported series of MDS/MPN with neutrophilia patients, conditioning intensity had no impact on the outcome (MAC were used in 76%). Noteworthy, an improved outcome following a combined fractionated 6–8 Gray TBI/FLU-conditioning regimen was reported in advanced CMML (Radujkovic et al. 2017).

In general, for young patients (<60 years), with a HCT-CI (Sorror et al. 2005) less than 3, MAC regimens such as BU-CY, TT/BU/FLU (TBF), TT/TREO/FLU (TTF), or the reduced-toxicity FLU/BUx4 (FB4) may be advisable, particularly in the proliferative variant of CMML and in other MDS/MPN with predominant proliferative features, whereas a RIC regimen such as BU/FLU, TREO10/FLU, or reduced TBF/TTF may be preferred for patients with older age or comorbidities and for patients undergoing transplant with disease remission following pretransplant treatment.

76.8 Maintenance/Post Transplant Strategies

As disease recurrence represents the major cause of transplant failure in MDS/MPN, there is a growing interest toward post transplant strategies, although few data are currently available in this particular setting.

Indirect evidence of a graft versus CMML by a reduced incidence of relapse in patients with chronic GvHD has been recently reported (Itonaga et al. 2018). Some effect of DLI has also been reported in patients with relapsing CMML and low disease burden.

With more molecular markers potentially available, cell therapy-based interventions may be planned on the base of residual or increasing MRD.

Potential interest both as preemptive and as maintenance strategy derive from the use of post transplant HMAs, alone or in combination with DLI, as reported in AML and MDS.

The use of lenalidomide and checkpoint inhibitors, but also JAK2 or PARP inhibitors, alone or even in combination, together with post transplant targeted therapies represents areas of growing interest under development.

Key Points

- MDS/MPNs include heterogeneous hematologic malignancies with mostly dismal prognosis (except low-risk CMML and MDS/MPN-SF3B1-T), for which allogeneic HCT still represents the only curative treatment option.
- In the absence of major comorbidities, allo-HCT is recommended for patients with high-risk CMML according to the CPSS or CPSS-Mol, MDS/MPN with neutrophila, and MDS/MPN-NOS up to 75 years of age.

- For young patients with a HCT-CI less than 3, MAC regimens may be advisable, particularly in the proliferative variant of CMML and in other MDS/MPN with predominant proliferative features, whereas a RIC regimen may be preferred for patients with older age or comorbidities.
- Although there is no evidence from dedicated trial, in patients with CMML-2 or other MDS/MPN with marrow blast higher than 10%, a pretransplant treatment based on HMAs or AML-like induction chemotherapy may be discussed.

References

Arber DA, Orazi A, Hasserjian R, et al. International consensus classification of myeloid neoplasms and acute leukemias: integrating morphologic, clinical, and genomic data. Blood. 2022;140(11):1200–28.

Assi R, Kantarjian HM, Garcia-Manero G, et al. A phase II trial of ruxolitinib in combination with azacytidine in myelodysplastic syndrome/myeloproliferative neoplasms. Am J Hematol. 2018;93:277–85.

Bennett JM, Catovsky D, Daniel MT, et al. The chronic myeloid leukaemias: guidelines for distinguishing chronic granulocytic, atypical chronic myeloid, and chronic myelomonocytic leukaemia. Proposals by the French-American-British cooperative Leukaemia group. Br J Haematol. 1994;87(4):746–54.

Benzarti S, Dskalakis M, Feller A, et al. Trends of incidence and survival of patients with chronic myelomonocytic. Leukemia between 1999 and 2014: a comparison between Swiss and American population-based cancer registries. Cancer Epidemiol. 2019;59:51–7.

Dao KHT, Tyner JW, Gotlib J. Recent Progress in chronic neutrophilic leukemia and atypical chronic myeloid leukemia. Curr Hematol Malig Rep. 2017;12:432–41.

Dao KHT, Gotlib J, Deininger MMN, et al. Efficacy of Ruxolitinib in patients with chronic neutrophilic leukemia and atypical chronic myeloid leukemia. J Clin Oncol. 2020;38(10):1006–18.

de Witte T, Bowen D, Robin M, et al. Allogeneic hematopoietic stem cell transplantation for MDS and CMML: recommendations from an international expert panel. Blood. 2017;129:1753–62.

DiNardo CD, Daver N, Jain N, et al. Myelodysplastic/myeloproliferative neoplasms, unclassifiable (MDS/MPN, U): natural history and clinical outcome by treatment strategy. Leukemia. 2014;28:958–61.

Eissa H, Gooley TA, Sorror ML, et al. Allogeneic hematopoietic cell transplantation for chronic myelomonocytic leukemia: relapse-free survival is determined by karyotype and comorbidities. Biol Blood Marrow Transplant. 2011;17:908–15.

Elena C, Gall A, Such E, et al. Integrating clinical features and genetic lesions in the risk assessment of patients with chronic myelomonocytic leukemia. Blood. 2016;128:1408–17.

Fontana D, Ramazzotti D, Aroldi A, et al. Integrated genomic, functional, and prognostic characterization of atypical chronic myeloid leukemia. Hema. 2020 Dec;4(6):e497.

Itonaga H, Aoki K, Aoki J, et al. Prognostic impact of donor source on allogeneic hematopoietic stem cell transplantation outcomes in adults with chronic myelomonocytic leukemia: a Nationwide retrospective analysis in Japan. Biol Blood Marrow Transplant. 2018;24:840–8.

Itzykson R, Kosmider O, Renneville A, et al. Prognostic score including gene mutations in chronic myelomonocytic leukemia. J Clin Oncol. 2013;31:2428–36.

Itzykson R, Fenaux P, Bowen D, et al. Diagnosis and treatment of chronic myelomonocytic leukemias in adults: recommendations from the European Hematology Association and the European LeukemiaNet. Hemasphere. 2018;2(6):e150.

Khoury JD, Solary E, Abla O, et al. The 5th edition of the world health organisation classification of haematolymphoids tumours: myeloid and histiocytic/dentritic neoplasms. Leukemia. 2022;36:1703–19.

Kongtim P, Popat U, Jimenez A, et al. Treatment with hypomethylating agents before allogeneic stem cell transplant improves progression free survival for patients with chronic myelomonocytic leukemia. Biol Blood Marrow Transplant. 2016;22:47–53.

Kröger N, Zabelina T, Guardiola P, et al. Allogeneic stem cell transplantation of adult chronic myelomonocytic leukaemia. A report on behalf of the chronic Leukaemia working Party of the European Group for blood and marrow transplantation (EBMT). Br J Haematol. 2002;118:67–73.

Kuendgen A, Kasprzak A, Germing U. Hybrid or mixed myelodysplastic/myeloproliferative disorders—epidemiological features and overview. Front Oncol. 2021;11:778741.

Liu HD, Ahn KW, Hu ZH, et al. Allogeneic hematopoietic cell transplantation for adult chronic myelomonocytic leukemia. Biol Blood Marrow Transplant. 2017;23:767–75.

Mangaonkar AA, Swoboda DM, Coltro G, et al. Clinicopathologic characteristics, prognostication and treatment outcomes for myelodysplastic/myeloproliferative neoplasm, unclassifiable (MDS/MPN-U): Mayo Clinic-Moffitt Cancer Center study of 135 consecutive patient. Leukemia. 2020;34(2):656–61.

Odenike O, Onida F, Padron E. Myelodysplastic syndromes and myelodysplastic/myeloproliferative

neoplasms: an update on risk stratification, molecular genetics, and therapeutic approaches including allogeneic hematopoietic stem cell transplantation. Am Soc Clin Oncol Educ Book. 2015;35:e398–412.

Onida F. Models of prognostication in chronic myelomonocytic leukemia. Curr Hematol Malig Rep. 2017;12:513–21.

Onida F, Beran M. Diagnosis and management of chronic myelomonocytic leukemia. Curr Hematol Malig Rep. 2008;3:31–6.

Onida F, Ball G, Kantarjian HM, et al. Characteristics and outcome of patients with Philadelphia chromosome negative, bcr/abl negative chronic myelogenous leukemia. Cancer. 2002;95:1673–84.

Onida F, Barosi G, Leone G, et al. Management recommendations for chronic myelomonocytic leukemia: consensus statements from the SIE, SIES. GITMO groups Haematologica. 2013;98:1344–52.

Onida F, de Wreede LC, van Biezen A, et al. Allogeneic stem cell transplantation in patients with atypical chronic myeloid leukaemia: a retrospective study from the chronic malignancies working Party of the European Society for blood and marrow transplantation. Br J Haematol. 2017;177:759–65.

Park S, Labopin M, Yakoub-Agha I, et al. Allogeneic stem cell transplantation for chronic myelomonocytic leukemia: a report from the Societe Francaise de Greffe de Moelle et de Therapie Cellulaire. Eur J Haematol. 2013;90:355–64.

Patnaik MM, How I. Diagnose and treat chronic myelomonocytic leukemia. Haematologica. 2022;107(7):1503–17.

Patnaik MM, Tefferi A. Refractory anemia with ring Sideroblasts (RARS) and RARS with thrombocytosis (RARS-T) – "2017 update on diagnosis, risk-stratification, and management". Am J Hematol. 2017;92(3):297–310.

Patnaik MM, Tefferi A. chronic myelomonocytic leukemia: 2022 update on diagnosis, risk stratification, and management. Am J Hematol. 2022;97(3):352–72.

Patnaik MM, Tefferi A. Atypical chronic myeloid leukemia and myelodysplastic/myeloproliferative neoplasm, not otherwise specified: 2023 update on diagnosis, risk stratification, and management. Am J Hematol. 2023;98(4):681–9.

Patnaik MM, Itzykson R, Lasho T, et al. ASXL1 and SETBP1 mutations and their prognostic contribution in chronic myelomonocytic leukemia: a two-center study of 466 patients. Leukemia. 2014;28:2206–12.

Patnaik MM, Wassie EA, Padron E, et al. Chronic myelomonocytic leukemia in younger patients: molecular and cytogenetic predictors of survival and treatment outcome. Blood Cancer J. 2015;5:e280.

Patnaik MM, Lasho TL, Finke CM, et al. Predictors of survival in refractory anemia with ring sideroblasts and thrombocytosis (RARS-T) and the role of next-generation sequencing. Am J Hematol. 2016;91:492–8.

Patnaik MM, Barraco D, Lasho TL, et al. Targeted next generation sequencing and identification of risk factors in World Health Organization defined atypical chronic myeloid leukemia. Am J Hematol. 2017;92(6):542–8.

Pophali P, Matin A, Mangaonkar AA, et al. Prognostic impact and timing considerations for allogeneic hematopoietic stem cell transplantation in chronic myelomonocytic leukemia. Blood Cancer J. 2020;10:121.

Radujkovic A, Hegenbart U, Müller-Tidow C, et al. High progression-free survival after reduced intensity total body irradiation-based conditioning in patients allografted for chronic myelomonocytic leukemia (CMML). Blood. 2017;130:4571.

Renneville A, Patnaik MM, Chan O, et al. Increasing recognition and emerging therapies argue for dedicated clinical trials in chronic myelomonocytic leukemia. Leukemia. 2021;35(10):2739–51.

Robin M, de Wreede LC, Padron E, et al. Role of allogeneic transplantation in chronic myelomonocytic leukemia: an international collaborative analysis. Blood. 2022;140(12):1408–18.

Rovó A, Gras L, Piebenbroek B et al. Outcomes of CMML patients undergoing allo-HCT are significantly worse compared to MDS - A study of the CMWP of the EBMT. American Journal of Hematology. 2023. In press.

Savona MR, Malcovati L, Komrokji R, et al. An international consortium proposal of uniform response criteria for myelodysplastic/myeloproliferative neoplasms (MDS/MPN) in adults. Blood. 2015;125:1857–65.

Sharma P, Shinde SS, Damlaj M, et al. Allogeneic hematopoietic stem cell transplant in adult patients with myelodysplastic syndrome/myeloproliferative neoplasm (MDS/MPN) overlap syndromes. Leuk Lymphoma. 2017;58:872–81.

Sorror ML, Maris MB, Storb R, et al. Hematopoietic cell transplantation (HCT)-specific comorbidity index: a new tool for risk assessment before allogeneic HCT. Blood. 2005;106:2912–9.

Such E, Germing U, Malcovati L, et al. Development and validation of a prognostic scoring system for patients with chronic myelomonocytic leukemia. Blood. 2013;121:3005–15.

Sun Y, Wang Q, Zhang X, et al. Molecular genetics and management of world health organization defined atypical chronic myeloid leukemia. Ann Hematol. 2023;102:777–85.

Symeonidis A, van Biezen A, de Wreede L, et al. Achievement of complete remission predicts outcome of allogeneic haematopoietic stem cell transplantation in patients with chronic myelomonocytic leukaemia. A study of the chronic malignancies working Party of the European Group for blood and marrow transplantation. Br J Haematol. 2015;171:239–46.

Tong X, Li J, Zhou Z, Zheng D, Liu J, Su C. Efficacy and side-effects of decitabine in treatment of atypical chronic myeloid leukemia. Leuk Lymphoma. 2015;56(6):1911–3.

Wang SA, Hasserjian RP, Fox PS, et al. Atypical chronic myeloid leukemia is clinically distinct from unclassifiable myelodysplastic/myeloproliferative neoplasms. Blood. 2014;123:2645–51.

Open Access This chapter is licensed under the terms of the Creative Commons Attribution 4.0 International License (http://creativecommons.org/licenses/by/4.0/), which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license and indicate if changes were made.

The images or other third party material in this chapter are included in the chapter's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the chapter's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

